

Mikkel Moth Billing - Aflevering 2

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1. Show that for every $k \geq 0$ the functions

$$\cos(k\frac{\pi t}{3}), \sin(k\frac{\pi t}{3}) : \mathbb{R} \rightarrow \mathbb{R}$$

are indeed elements of $L^2([-3, 3])$.

We show this with def. 0.4

$$\begin{aligned} \int_{-3}^3 |\cos(k\frac{\pi t}{3})|^2 dt &= \int_{-3}^3 \left(\frac{1}{2} + \frac{1}{2} \cos(2k\frac{\pi t}{3}) \right) dt \\ &= \left[\frac{1}{2}t + \frac{3}{4} \frac{1}{k\pi} \sin(2k\frac{\pi t}{3}) \right]_{-3}^3 \\ &= \frac{3}{2} + \frac{3}{4} \frac{1}{k\pi} \sin(2k\pi) + \frac{3}{2} - \frac{3}{4} \frac{1}{k\pi} \sin(-2k\pi) \\ &= 3 < \infty \end{aligned}$$

$$\begin{aligned} \int_{-3}^3 |\sin(k\frac{\pi t}{3})|^2 dt &= \int_{-3}^3 \left(\frac{1}{2} - \frac{1}{2} \cos(2k\frac{\pi t}{3}) \right) dt \\ &= \left[\frac{1}{2}t - \frac{3}{4} \frac{1}{k\pi} \sin(2k\frac{\pi t}{3}) \right]_{-3}^3 \\ &= 3 < \infty \end{aligned}$$

So the functions are elements in $L^2([-3, 3])$

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2. Are the functions $\cos(k\frac{\pi t}{3}), \sin(k\frac{\pi t}{3}) : \mathbb{R} \rightarrow \mathbb{R}$ periodic, for each $k \in \mathbb{N}$? If yes, what is their period?

To be periodic a function must satisfy $f(x+p) = f(x)$, where p is the period

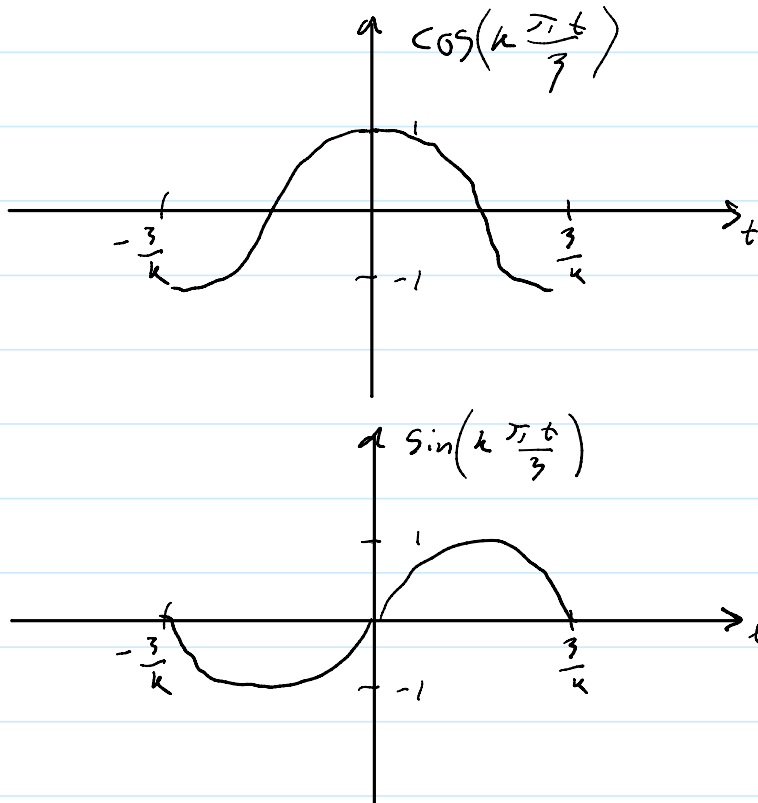
We know that $h(x) = f(ax)$ is periodic with period $\frac{p}{a}$, if $f(x)$ has period p .

We know $\cos(t)$ and $\sin(t)$ are 2π periodic therefore $\cos(\frac{k\pi}{3}t)$ and $\sin(\frac{k\pi}{3}t)$ must have period

$$\frac{p}{a} = \frac{2\pi}{\frac{k\pi}{3}} = \frac{6}{k}.$$

3. Which of the functions $\cos(k\frac{\pi t}{3}), \sin(k\frac{\pi t}{3}) : \mathbb{R} \rightarrow \mathbb{R}$ are odd/even?

We can show this by drawing the functions:



We can see that $\cos(k \frac{\pi t}{3})$ is even
and $\sin(k \frac{\pi t}{3})$ is odd.

4. Show that:

$$\frac{1}{3} \int_{-3}^3 \cos(k \frac{\pi t}{3}) \cos(j \frac{\pi t}{3}) dt = \begin{cases} 1 & \text{if } k = j \geq 1 \\ 2 & \text{if } k = j = 0 \\ 0 & \text{if } k \neq j. \end{cases}$$

$$\frac{1}{3} \int_{-3}^3 \sin(k \frac{\pi t}{3}) \sin(j \frac{\pi t}{3}) dt = \begin{cases} 1 & \text{if } k = j \geq 1 \\ 0 & \text{if } k \neq j. \end{cases}$$

$$\frac{1}{3} \int_{-3}^3 \cos(k \frac{\pi t}{3}) \sin(j \frac{\pi t}{3}) dt = 0, \text{ for all integers } k, j.$$

From Theorem 1.1 and the lectures,
we know that $\cos(kx)$ and $\sin(kx)$,
which are elements in $L^2([-\pi, \pi])$,
convert to $\cos(k \frac{\pi t}{a})$ and $\sin(k \frac{\pi t}{a})$,
which are elements in $L^2([-a, a])$
for $a \in \mathbb{R}$ $a > 0$.

If $a=3$, we can substitute π with 3
and nx with $\frac{k\pi t}{3}$ and kx with $\frac{j\pi t}{3}$
in Theorem 1.1 we get the answer.

5. Compute

$$\langle \cos(k \frac{\pi t}{3}), \sin(j \frac{\pi t}{3}) \rangle_{L^2([-3, 3])} = ?$$

$$\langle \cos(k \frac{\pi t}{3}), \cos(j \frac{\pi t}{3}) \rangle_{L^2([-3, 3])} = ?$$

$$\langle \sin(k \frac{\pi t}{3}), \sin(j \frac{\pi t}{3}) \rangle_{L^2([-3, 3])} = ?$$

We can use exercise 4 to calculate this.

$$\langle \cos(k \frac{\pi t}{3}), \sin(j \frac{\pi t}{3}) \rangle = \int_{-3}^3 \cos(k \frac{\pi t}{3}) \sin(j \frac{\pi t}{3}) dt \quad P. 3/6$$

$$= 3 \cdot \frac{1}{3} \int_{-3}^3 \cos\left(k \frac{\pi t}{3}\right) \sin\left(j \frac{\pi t}{3}\right) dt$$

= 0 for all integers k and j

$$\begin{aligned} \left\langle \cos\left(k \frac{\pi t}{3}\right), \cos\left(j \frac{\pi t}{3}\right) \right\rangle &= 3 \cdot \frac{1}{3} \int_{-3}^3 \cos\left(k \frac{\pi t}{3}\right) \cos\left(j \frac{\pi t}{3}\right) dt \\ &= \begin{cases} 3 & \text{if } k=j \geq 1 \\ 6 & \text{if } k=j=0 \\ 0 & \text{if } k \neq j \end{cases} \end{aligned}$$

$$\begin{aligned} \left\langle \sin\left(k \frac{\pi t}{3}\right), \sin\left(j \frac{\pi t}{3}\right) \right\rangle &= 3 \cdot \frac{1}{3} \int_{-3}^3 \sin\left(k \frac{\pi t}{3}\right) \sin\left(j \frac{\pi t}{3}\right) dt \\ &= \begin{cases} 3 & \text{if } k=j \geq 1 \\ 0 & \text{if } k \neq j \end{cases} \end{aligned}$$

6. Compute the real Fourier coefficients of the function $f(t) = t^3$, defined on the interval $[-3, 3]$, with respect to the inner product space $L^2([-3, 3])$. Then write the real Fourier series associated with f and the space $L^2([-3, 3])$.

We follow Theorem 1.4 with $a=3$
and use that t^3 is an odd function.

$$a_0 = \frac{1}{6} \int_{-3}^3 t^3 dt = 0$$

$$a_n = \frac{1}{3} \int_{-3}^3 t^3 \cos\left(\frac{n\pi t}{3}\right) dt = 0 : \text{odd} \cdot \text{even} = \text{odd}$$

$$b_n = \frac{1}{3} \int_{-3}^3 t^3 \sin\left(\frac{n\pi t}{3}\right) dt : \text{odd} \cdot \text{odd} = \text{even}$$

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$$= \frac{2}{3} \int_0^3 t^3 \sin\left(\frac{n\pi t}{3}\right) dt \quad \text{integration by parts:}$$

$$= \frac{2}{3} \left(\frac{3^3 \cdot 3 \cdot (-\cos(n\pi))}{n\pi} + \frac{3^2}{n\pi} \int_0^3 t^2 \cos\left(\frac{n\pi t}{3}\right) dt \right)$$

$$= \frac{2}{3} \left(-\frac{3^4}{n\pi} (-1)^n + \frac{3^2}{n\pi} \left(3^2 \overset{0}{\cancel{\sin(n\pi)}} - \frac{2 \cdot 3}{n\pi} \int_0^3 t \sin\left(\frac{n\pi t}{3}\right) dt \right) \right)$$

$$= \frac{2}{3} \left(-\frac{3^4}{n\pi} (-1)^n - \frac{2 \cdot 3^3}{n^2 \pi^2} \cdot \left(-\frac{2}{3} \frac{\cos(n\pi)}{n\pi} + \frac{2}{n\pi} \int_0^3 \cos\left(\frac{n\pi t}{3}\right) dt \right) \right)$$

$$= \frac{2}{3} \cdot \left(-\frac{3^4}{n\pi} (-1)^n + \frac{2 \cdot 3^4}{n^3 \pi^3} \left(3 \cdot (-1)^n - \overset{0}{\cancel{\sin(n\pi)}} \right) \right)$$

$$= \frac{2 \cdot 2 \cdot 3^4}{n^3 \pi^3} (-1)^n - \frac{2 \cdot 3^3}{n^2 \pi^2} (-1)^n$$

$$= \frac{2 \cdot 3^3}{n^3 \pi^3} (-1)^n (2 \cdot 3 - n^2 \pi^2)$$

$$= \frac{54}{n^3 \pi^3} (-1)^n (6 - n^2 \pi^2)$$

Therefore:

$$F(x) = \sum_{n=1}^{\infty} \frac{54}{n^3 \pi^3} (-1)^n (6 - n^2 \pi^2) \sin\left(\frac{n\pi x}{2}\right)$$

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7. Re-write the real Fourier series associated with f as a complex Fourier series with respect to $L^2([-3, 3])$.

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$$\alpha_n = \frac{1}{2}(a_n - ib_n), \quad a_n = 0$$
$$= -i \frac{27}{n^3 \pi^3} (-1)^n (6 - n^2 \pi^2)$$

$$\Rightarrow F(x) = \sum_{n=-\infty}^{\infty} \alpha_n e^{inx} = \sum_{n=-\infty}^{\infty} -i \frac{27}{n^3 \pi^3} (-1)^n (6 - n^2 \pi^2) e^{inx}$$

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